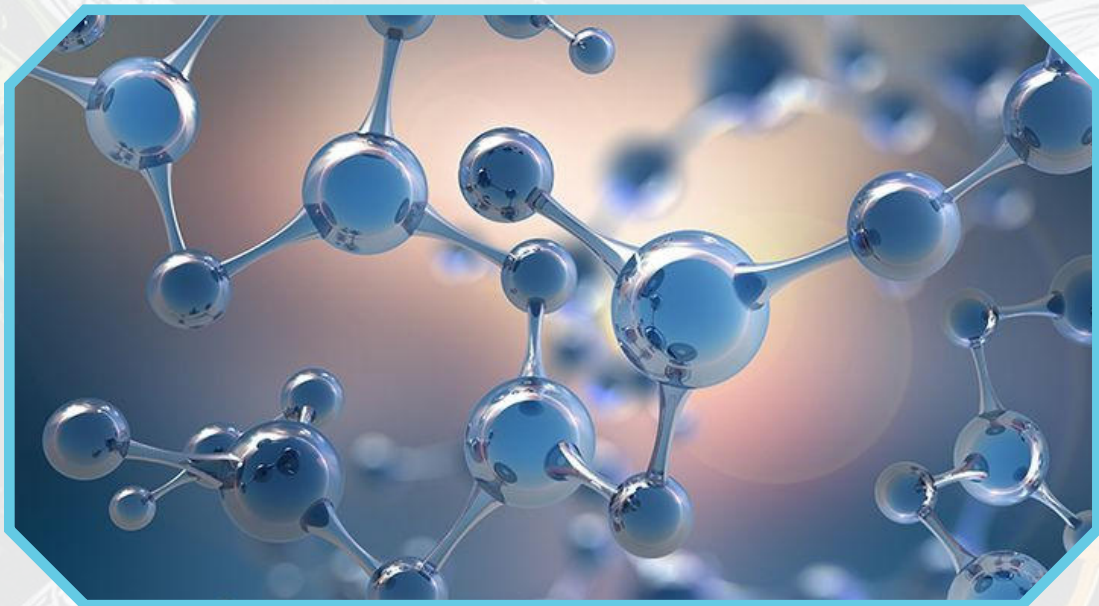




SOLID-STATE OF ELECTRONICS



Solid-state electronics is a branch of electronics that deals with the flow of electric current through solid materials, such as semiconductors, insulators, and conductors.

It involves the design, development, and application of electronic devices and circuits that use these materials to control and manipulate electrical energy.

Solid-state electronics has revolutionized the field of electronics, enabling the creation of smaller, faster, and more efficient devices, such as transistors, diodes, integrated circuits, and microprocessors, which are essential components in modern computers, smart phones, and other electronic devices.

The field is constantly evolving, with advancements in materials science, nanotechnology, and quantum mechanics.

Intrinsic (Pure) and Extrinsic (Doped) Semiconductors:

Intrinsic (Pure) Semiconductors:

Intrinsic semiconductors are materials that are chemically pure, meaning they do not contain any significant amount of impurity atoms.

The two most commonly used intrinsic semiconductors are **silicon (Si)** and **germanium (Ge)**.

These materials have a crystalline structure, and their electrical properties are determined by the arrangement and behavior of their atoms.

Extrinsic (Doped) Semiconductors:

Doped semiconductors are created by introducing impurity atoms into the intrinsic semiconductor material. This process is known as doping, and it significantly alters the electrical properties of the semiconductor.

Production of P-Type and N-Type Semiconductors:

The intentional addition of impurities alters the conductivity and electrical properties of the semiconductor, allowing for the controlled operation of charge carriers.

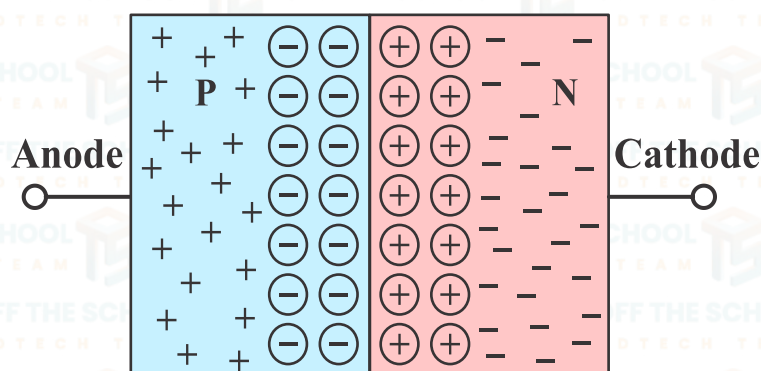
N-type Semiconductors: electron

It is characterized by an excess of negatively charged carriers, known as "electrons," or majority negative charge carriers.

P-type Semiconductors:

P-N Junction:

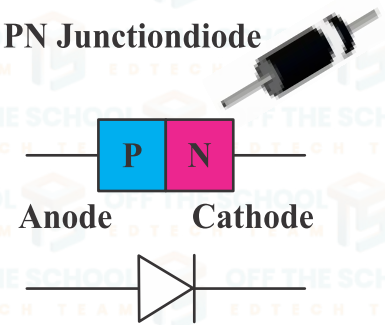
The P-N junction is formed when the p-type and n-type semiconductors are joined, and is called the P-N junction.



P-N Junction as Diode:

A diode is a semiconductor device that is formed through the P-N junction and is used in allowing the flow of electric current in one direction and blocking it in the opposite.

PN Junction diode



Properties of Diode:

Below are some of the common properties of a diode:

1. Diode has the ability to rectify electric current.
2. It can create a potential barrier and make use of its capacitance properties.
3. Diode creates various nonlinear current-voltage characteristics.

Diode Biasing:

Forward Bias:

When the P-type is connected to the positive terminal of the battery and the N-type is connected to the negative terminal, it is called **Forward bias**.

Reverse Bias:

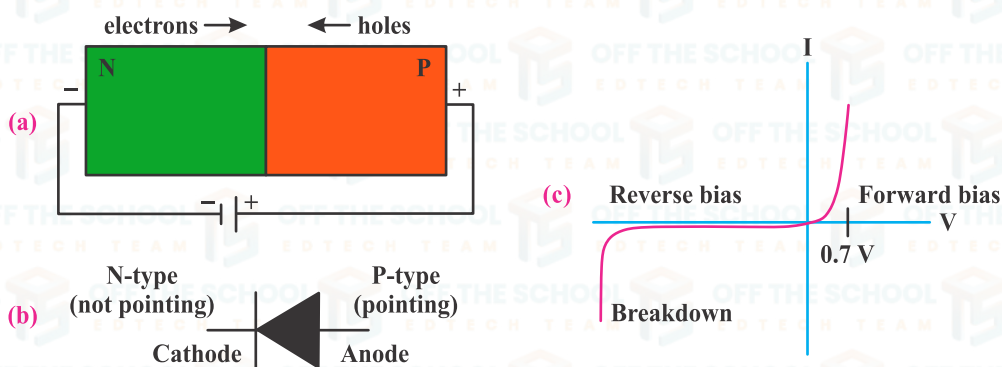
When the P-type is connected to the N-type negative terminal of the battery and the N-type is connected to the positive side, it is called **Reverse bias**.

I-V Characteristics of p-n Junction:

The relationship between the voltage across the junction and current through the circuit is known as the (V-I) characteristics of a P-N junction or semiconductor diode.

The V-I characteristics of the P-N junction can be explained in three cases:

- Zero bias or unbiased
- Forward bias
- Reverse bias



Rectification:

Rectification is the process of converting an alternating current (AC) waveform into a direct current (DC) waveform, i.e., creating a new waveform that has only a single polarity.

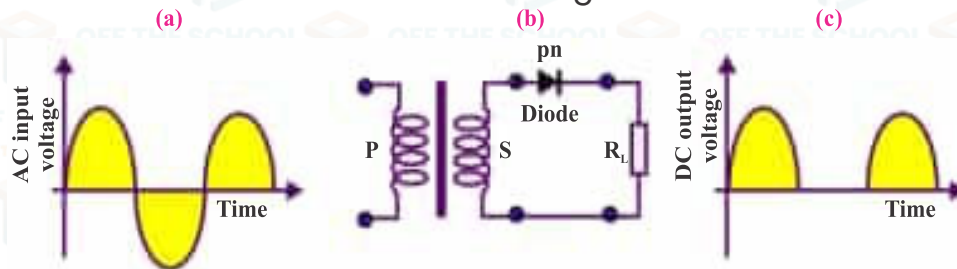


Rectification is classified into two types according to the output characteristics, which are:

- (i) half-wave rectification and
- (ii) full-wave rectification.

Half Wave Rectification:

Since a diode allows AC current to flow only in one direction, it can serve as a rectifier. The AC source applies a voltage across the diode alternately positive and negative. When the positive cycle of AC voltage passes through the diode, the diode is forward biased and acts as a closed circuit. There is current through the resistor R .



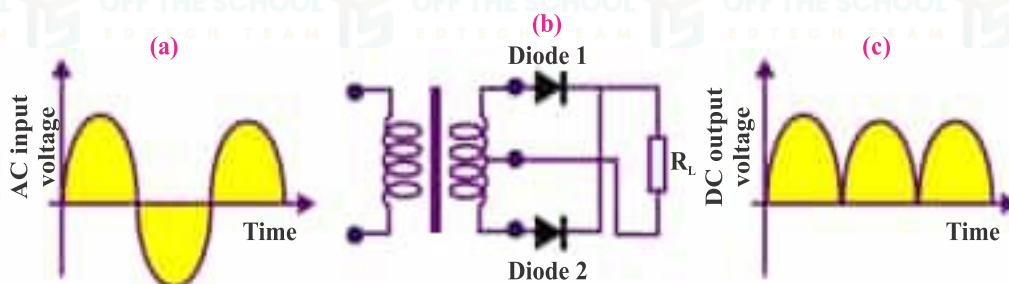
Applications of a Half-wave Rectifier:

- Low power simple battery charger circuit.
- Fire alarm circuits.
- Soldering iron circuit.

Full-Wave Rectification:

Full-wave rectification uses two diodes. During the positive half cycle of input AC signal, this makes the diode D_1 forward biased (acts as a closed switch) and diode D_2 reverse biased (acts as an open switch). Therefore, current flows through the load resistor R .

During the negative half cycle of input AC signal, this makes the diode D_1 reverse biased and the diode D_2 forward biased. Therefore, the current will flow through diode D_2 through the load resistor R and lower half of the secondary winding.



Applications of a full-wave rectification:

- Mobile phones, laptops, charger circuits.
- Uninterruptible Power Supply (UPS) circuits to convert AC to DC.
- Our home inverters convert AC to DC.
- LCD, LED TVs.

Function and uses of light emitting diode (LED), Photodiode, Photo Voltaic cell:

1. Light Emitting Diode (LED):

A light-emitting diode (LED) is a special type of junction in which current flows when it is activated in a forward direction.

2. Photodiode:

A photodiode is a type of light detector that converts light into current or voltage. It includes optical filters, built-in lenses, and surface areas.

3. Photo Voltaic Cells:

Solar cell or photovoltaic cell is heavily doped P-N junction diodes used to convert sunlight into electric energy.

Uses of Photodiodes:

- **Camera:** Light meters, Automatic Shutter Control, Photographic Flash Control.
- **Medical:** CAT Scanners, X-Ray Detection, Pulse Oximeters, Blood Particle Analysis.
- **Automotive:** Headlight Dimmer, Twilight Detectors.
- **Communication:** Fiber Optic Link, Optical Remote Control.
- **Industry:** Bar Code Scanners, Light Pen, Encoders, Surveying Instruments, Copiers - Density of Toner.

Transistors and their characteristics:

A transistor is a semiconductor device that amplifies or switches electronic signals by controlling the flow of current between its three layers: base, collector, and emitter. It acts as a gatekeeper, allowing or blocking the flow of current between the collector and emitter, based on the voltage applied to the base.

Both **N-P-N** and **P-N-P** transistors can be made.

There are three main characteristics listed below:

1. Input Characteristics
2. Output Characteristics
3. Constant-Current Characteristics.

Transistor's current equation:

It is a three-terminal device, with three possible ways to connect it within an electronic circuit with one terminal being common to both the input and output signals.

Bipolar Junction Transistor Configurations:

Each method of connection responds differently to its input signal within a circuit as the static characteristics of the transistor vary with each circuit arrangement.

The Common Emitter Amplifier Circuit:

In this circuit shown, the current leaving the transistor must be equal to the currents entering it because the emitter current (I_E) is the sum of the collector current (I_C) and the base current (I_B).

The common emitter transistor configuration has a substantial current gain because the load resistance (R_L) is connected in series with the collector. The current gain is represented by Beta (β), which is the ratio of I_C to I_B . For the common emitter setup:

$$I_E = I_C + I_B$$

where the ratio of I_C to I_E is known as Alpha (α). It's important to note that Alpha is always less than one.

The mathematical relationship between Alpha (α) and Beta (β) expresses the current gain of the transistor.

$$\alpha = I_C / I_E$$

$$\beta = I_C / I_B$$

$$\text{As } \alpha = \beta / (\beta + 1) \text{ and } \beta = \alpha / (1 - \alpha)$$

$$I_E = I_C + I_B$$

Where:

- **IC** is the current flowing into the collector terminal.
- **IB** is the current flowing into the base terminal.
- **IE** is the current flowing out of the emitter terminal.

Operation Principles and -V Characteristics and Methods of Transistor Biasing:

Bipolar Junction Transistor (BJT):

Operation Principles:

1. Emitter, Base, and Collector:

- The three layers of a BJT are called the emitter (E), base (B), and collector (C).
- The flow of current in an N-P-N transistor is from the emitter to the collector, and in a P-N-P transistor, it's from the collector to the emitter.

2. Transistor Action:

The operation of a BJT involves the injection of minority carriers (electrons in N-P-N or holes in P-N-P) from the emitter into the base region. This controls the majority carrier flow from the collector to the emitter.

Biasing Methods:

Biasing is the process of applying external voltages to a transistor to establish a desired operating point for proper amplification or switching.

There are several biasing methods for transistors. Here are some common biasing methods:

Bipolar Junction Transistor (BJT) Biasing Methods:

1. Fixed Bias (Base Bias):

- Connects a resistor between the base and the power supply to establish a fixed voltage at the base.
- Simple but not very stable.

2. Emitter Bias (Emitter Stabilized Bias):

- Connects a resistor from the emitter to the power supply, stabilizing the operating point.
- More stable than fixed bias.

3. Collector Feedback Bias:

- Combines features of fixed bias and emitter bias for improved stability.
- Uses a resistor network to provide feedback.

4. Voltage Divider Bias:

- Utilizes a resistor divider network to bias the base and provide stability.
- More stable than fixed bias but less stable than emitter bias.

The transistors as a switch:

A transistor can be used as a switch in electronic circuits, where it functions to either allow or block the flow of current.

There are two main configurations for using a transistor as a switch: the NPN (negative-positive-negative) and PNP (positive-negative-positive) configurations.

NPN Transistor as a Switch:

Here are key points to note:

1. Off State:

- Normally, Q1 allows no output current unless we apply forward voltage to its base-emitter circuit.

2. Forward Voltage:

The amount of output current is controlled by the forward voltage that controls the base current.

- The input control circuit determines base current.
- The output current is collector current for the power circuit.
- Q1 is an NPN transistor, needing a positive V_{BE} for forward voltage.
- The emitter is common to both input control and power circuits.
- Common-emitter (CE) circuit is the most common transistor arrangement.

When switch S1 is open:

- No current flows the base-emitter or control circuit because no forward voltage is applied.
- Resistance from emitter to collector of the transistor is very high.
- No current flows in the power circuit, and the lamp doesn't light up.
- A small current flows in the control circuit.
- R1 limits current in the base circuit.
- Resistance from emitter to collector of Q1 decreases.
- A large current flows in the power circuit, lighting up the lamp.

Opening switch in the control circuit turns off the lamp in the power circuit because resistance from emitter to collector of Q1 increases again, almost to infinity.

Common Collector (CC) Circuit:

Advantages:

- CC amplifiers offer high current gain and low input impedance, making them suitable for applications where a high-resistance input needs to drive a low-resistance output load.

Applications:

- Commonly used in impedance matching and signal buffering circuits due to their favorable characteristics.

Operational Amplifier:

An operational amplifier is a direct-coupled high gain amplifier. It can be operated on both AC and DC signals. Operational Amplifier, also called as an Op-Amp, is an integrated circuit, which can be used to perform various linear, non-linear, and mathematical operations.

Effects of Negative Feedback on the Gain of an Operational Amplifier:

Negative Feedback:

Negative feedback occurs when a portion of the output signal, like voltage or current, is feedback into the input in a way that opposes or subtracts from the input signal.

Advantages:

- 1. Reduced Gain:** Negative feedback decreases the overall gain of the system.
- 2. Decreased Distortion and Noise:** It minimizes distortion and noise, leading to a cleaner output signal.
- 3. Improved Stability:** The gain becomes more stable and less sensitive to external changes, such as temperature.
- 4. Increased Bandwidth:** Enhances the system's bandwidth.
- 5. Better Impedance:** Lowers the output impedance and raises the input impedance.